

Feature-Set Scaling for Kidney Failure Prediction: Benchmarking Survival Models Against the Kidney Failure Risk Equation on MIMIC-IV

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Abstract

The Kidney Failure Risk Equation (KFRE) predicts progression to end-stage renal disease (ESRD) from as few as four routinely collected variables, yet modern EHR resources and deep survival architectures raise the question of whether more features and more expressive models meaningfully improve on it. We isolate this question on a shared MIMIC-IV-derived cohort of chronic kidney disease patients by benchmarking ten survival models plus the published closed-form KFRE equation itself across three nested feature-set regimes reconstructing KFRE’s own 4- and 8-variable inputs plus a 20-feature heterogeneous stress test. Building KFRE-comparable cohorts required a new creatinine-anchored, bounded-window row-merge design for combining multiple asynchronously-drawn labs into simultaneous covariate snapshots. Once every model is evaluated on an identical, per-patient basis — a methodological correction that reverses an earlier, row-level-evaluated result in which KFRE and Cox appeared to lead — KFRE’s discrimination is matched by most learned models rather than clearly beaten by them, only one deep sequence model (Dynamic-DeepHit) shows a decisive advantage, and nearly every model exceeds a simple eGFR-threshold referral rule’s clinical net benefit. We report a Cox model performing below chance as an investigated, disclosed open question rather than smoothing it over.

Keywords: Survival analysis, chronic kidney disease, Kidney Failure Risk Equation, MIMIC-IV, benchmark

Data and Code Availability This work uses MIMIC-IV (Johnson et al., 2023), available via PhysioNet to credentialed users under its data use agreement. Anonymized code implementing the cohort extraction, models, and analyses is provided as supple-

mentary material for review; a public repository will be released upon publication.

Institutional Review Board (IRB) No IRB approval was required. MIMIC-IV is a de-identified, publicly available critical care database released under a data use agreement; this study involved no additional identifiable patient contact.

1. Introduction

Chronic kidney disease (CKD) affects over 1 in 7 American adults and can progress to end-stage renal disease (ESRD) (Centers for Disease Control and Prevention, 2023). Early identification of high-risk patients is critical for nephrology referral and dialysis-access planning (Levin and Stevens, 2011). The clinical standard for this task is the Kidney Failure Risk Equation (KFRE) (Tangri et al., 2011), a closed-form Cox model built from as few as four routinely collected variables (age, sex, eGFR, urine albumin-to-creatinine ratio (uACR)), or eight in its extended form (adding calcium, phosphate, bicarbonate, albumin). KFRE has been externally validated across 31 cohorts spanning over 700,000 patients (Tangri et al., 2016) and is embedded in clinical guidelines (Major et al., 2019).

KFRE’s minimalism is a deliberate design choice, not a data limitation: it was derived under classical regression sample-size constraints from a handful of clinically available variables. Modern EHR resources and expressive deep survival architectures raise a question KFRE’s own literature does not answer: *does adding more labs and more flexible models meaningfully improve on it, or does the 4–8 variable KFRE feature set already capture most of the recoverable signal?* Existing CKD deep-learning benchmarks conflate this question, varying feature count and model architecture together.

We isolate it on a shared MIMIC-IV (John-son et al., 2023) CKD cohort, holding the population fixed and varying only the feature set across three nested regimes: **four_features** and **eight_features** (exact reconstructions of KFRE’s 4- and 8-variable inputs) and **twenty_features** (a heterogeneous 20-lab, missingness-flagged stress test with no published KFRE analogue), benchmarking 10 survival models per scenario plus the closed-form KFRE equation itself on the two KFRE-comparable scenarios (11 total there).

Building KFRE-comparable cohorts required a row-merge design not needed when every input is a single lab type: KFRE’s inputs are multiple, asynchronously-drawn labs that must be combined into simultaneous covariate snapshots without collapsing a patient’s repeated measurements into one row (Section 2.1).

Contributions. (1) A cohort-construction methodology for merging multiple asynchronously-drawn labs into KFRE-comparable, patient-level landmark rows, each design choice backed by prior clinical/statistical literature. (2) A direct benchmark of 10 survival models plus published KFRE across three feature-set sizes on the same cohort, with discrimination, calibration, and decision-curve analysis, aggregated across independent repetitions. (3) A methodological finding: evaluating time-varying survival models on raw per-visit rows rather than one prediction per patient inflates discrimination for several architectures, including KFRE itself — a pit-fall likely shared by other EHR time-to-event benchmarks reusing a counting-process row format. (4) A disclosed, investigated limitation: a Cox model performing below chance on this row format, reported openly rather than hidden.

2. Methods

Problem formulation. We frame progression to ESRD as right-censored time-to-event estimation. Each covariate row $\mathbf{x}_i \in \mathbb{R}^d$ is paired with a duration $t_i \in \mathbb{R}^+$ and event indicator $e_i \in \{0, 1\}$. Patient histories are stored in a counting-process (**start**, **stop**) format, supporting both single-landmark models (Cox, KFRE) and sequence models consuming a patient’s full row history (Dynamic-DeepHit, RNN-Surv).

	Source	four	eight	twenty
Patients	34,332	2,809	1,213	32,601
Records	74,197	26,829	5,407	8.13M
% male	57.2	58.3	60.4	57.1
Age, yrs (SD)	67.8 (16.1)	63.9 (15.1)	62.1 (15.4)	67.8 (16.1)
ESRD, %	91.9	83.5	85.1	91.5
Train/test pts	—	2247/562	970/243	26080/6521

Table 1: Cohort flow: source population vs. each scenario’s final cohort.

2.1. Cohort

The source population is 34,332 MIMIC-IV patients meeting a CKD stage 3–5 or ESRD diagnosis-code criterion (57.2% male, mean age 67.8, SD 16.1).

Four/eight_features (KFRE-comparable). KFRE’s inputs come from separate, asynchronously-ordered labs (creatinine, urine albumin/creatinine, and for the 8-variable form, a chemistry panel), so producing real, simultaneous, non-missing rows required a merge design (full derivation and literature citations in Appendix A): each row anchors on one creatinine draw (eGFR computed at that instant via CKD-EPI 2021); calcium/phosphate/bicarbonate/albumin are matched to the nearest value within ± 24 h (following APACHE II’s precedent for a concurrent chemistry-panel snapshot); uACR is matched to the nearest value across the patient’s whole history (loosened from a same-admission bound after that bound caused 96.5% cohort attrition with outcome-rate skew); rows missing a required lab are dropped, not imputed. This keeps four_features/eight_features complete-case and time-varying (one row per qualifying creatinine draw), matching KFRE’s own eligibility criteria.

Twenty_features (heterogeneous). We select the 20 most frequently measured labs for this cohort, each represented as a (value, missingness-flag) pair (40 columns), one row per lab event, no cross-lab merge — testing whether feature count alone, at realistic EHR sparsity, adds discriminative power.

Table 1 reports each scenario’s final cohort. eight_features’ 1,213 patients/1,032 events is the smallest cohort here but exceeds 9 of the 15 published cohorts (263–12,955 patients) the 8-variable KFRE was externally validated on (Tangri et al., 2016), with events-per-variable ≈ 129 — $13\times$ the classical $EPV \geq 10$ adequacy floor (Peduzzi et al., 1996).

2.2. KFRE Baseline

For four/eight_features we add KFRE as a non-trained benchmark: $\text{Risk}(t) = 1 - S_0(t)^{\exp(L)}$, North-American $S_0(2\text{yr}) = 0.9750$ (Tangri et al., 2016), with

$$L_4 = -0.2201\left(\frac{\text{age}}{10} - 7.036\right) + 0.2467(\text{male} - 0.5642) - 0.5567\left(\frac{\text{eGFR}}{5} - 7.222\right) + 0.4510(\ln(\text{uACR}) - 5.137) \quad (1)$$

for the 4-variable form and an analogous 8-term L_8 (adding albumin, phosphate, bicarbonate, calcium terms) for the 8-variable form (Tangri et al., 2011) (full coefficients in Appendix B).

2.3. Benchmarked Models

We benchmark 10 learned models per scenario, plus KFRE on four/eight_features only (11 total there): **Cox** (lifelines’ CoxTimeVaryingFitter, fit directly on the counting-process rows) and **Weibull AFT** (Cox, 1972; Anderson, 1991; Davidson-Pilon, 2019); the deep sequence models **DeepSurv** (Katzman et al., 2018), **Dynamic-DeepHit** (Lee et al., 2020, 2018), **RNN-Surv** (LSTM-based) (Giunchiglia et al., 2018), **LogisticHazard** (pycox’s off-the-shelf discrete-time hazard model), and **Hazard Transformer** (Oliver et al., 2024; Vaswani et al., 2017); and the ensembles **GBSA** (Chen et al., 2013), **Random Survival Forest** (Pickett et al., 2021; Ishwaran et al., 2008), and **FastSurvivalSVM**, via **scikit-survival** (Pölsterl, 2020). Every architecture’s exact structural form, loss, and tuned hyperparameter range — verified directly against the model implementations, not restated from a fixed reference configuration — is given in Appendix C; neural models are tuned via 10-trial Bayesian search (Optuna) per scenario, GBSA/SRF via grid search, and FastSurvivalSVM/Cox/Weibull AFT use fixed hyperparameters.

2.4. Evaluation

We report Harrell’s C-index, integrated Brier score, and mean time-dependent AUC at the 730-day (2-year) horizon. **Per-patient evaluation.** Three of the eleven models (Cox, RNN-Surv, KFRE) initially scored every raw time-varying row independently while the remaining models internally deduplicate to one prediction per patient; since a patient’s rows are highly correlated, row-level scoring both deflates Brier score and inflates C-index for those three

models. All results here score every model identically on one prediction per patient (each patient’s final observation at test time) — a correction that reverses an earlier row-level-evaluated pass in which KFRE and Cox appeared to outperform most learned models.

We additionally report calibration (predicted risk per decile vs. Kaplan-Meier-observed risk, at 2- and 5-year horizons) and decision-curve/net-benefit analysis (Vickers and Elkin, 2006) against treat-all, treat-none, and a non-model eGFR-threshold referral rule (eGFR < 30 or < 45). A competing-risks analysis (death before ESRD) was not included, as it would require re-deriving each subject’s absolute calendar anchor, not retained by the exported train/test files (Section 4.1).

3. Results

3.1. Discrimination: Four/Eight Features vs. KFRE

Table 2 reports C-index, integrated Brier, and time-dependent AUC at 730 days, evaluated per-patient (Section 2.4), aggregated as mean (SD) across repetitions.¹

Feature count barely moves most models. Nine of eleven models change by < 0.03 C-index between four_features and eight_features, including KFRE (0.539 on both); only Cox (+0.16) and Hazard Transformer (−0.06) move meaningfully, in opposite directions.

KFRE is not clearly beaten by most learned models. On equal per-patient footing, KFRE’s C-index lands squarely inside the cluster formed by Logistic Hazard, RNN-Surv, DeepSurv, GBSA, Survival RF, Survival SVM, and Weibull AFT (0.49–0.58) — none of which reliably outperforms the published clinical equation. Only **Dynamic-DeepHit** clearly beats it (C-index 0.83–0.87, AUC 0.94–0.95), consistent with being the only benchmarked model that consumes each patient’s full multi-visit sequence rather than a single landmark row.

Cox underperforms and remains an open question. Cox scores *below* the 0.5 random baseline on four_features (0.308) and near-random on eight_features (0.470) — the only model to do so

1. Values shown are from the aggregation pipeline (pkgs/scripts/aggregate_rep_metrics.py), which reports mean±SD across every completed repetition and is re-run as later repetitions finish.

Model	four_features		
	C-idx↑	Brier↓	AUC↑
Cox	0.308	0.531	0.272
Dynamic-DeepHit	0.832	0.285	0.947
Hazard Transformer	0.639	0.404	0.703
Logistic Hazard	0.552	0.416	0.571
RNN-Surv	0.516	0.449	0.519
DeepSurv	0.510	0.448	0.516
GBSA	0.519	0.433	0.535
Survival RF	0.531	0.445	0.541
Survival SVM	0.543	0.417	0.571
Weibull AFT	0.552	0.414	0.579
KFRE	0.539	0.552	0.553

Model	eight_features		
	C-idx↑	Brier↓	AUC↑
Cox	0.470	0.578	0.425
Dynamic-DeepHit	0.866	0.308	0.938
Hazard Transformer	0.576	0.442	0.608
Logistic Hazard	0.520	0.500	0.493
RNN-Surv	0.527	0.499	0.574
DeepSurv	0.494	0.495	0.510
GBSA	0.539	0.515	0.567
Survival RF	0.533	0.536	0.543
Survival SVM	0.540	0.488	0.595
Weibull AFT	0.554	0.485	0.597
KFRE	0.539	0.737	0.579

Table 2: Discrimination at the 2-year horizon, per-patient evaluation. Bold: best per column. KFRE is a closed-form clinical equation, not a trained model.

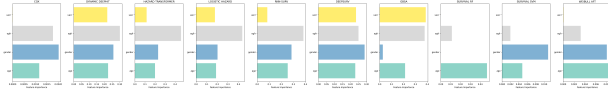


Figure 1: Per-model feature importance, four_features.

on either scenario, and not yet explained by an architectural cause we can point to (Section 4.1).

3.2. Feature Importance, Calibration, Decision Curves

Figure 1 shows each model’s native importance measure (Cox/Weibull/SVM: $|\text{coefficient}|$; GBSA: impurity-based; SRF: permutation importance; remaining neural models: mean $|\text{gradient}|$ — not SHAP values, despite this analysis’s inherited file naming). eGFR dominates for models that use it at all (0.25–0.46); uACR’s contribution is architecture-dependent, substantial for GBSA/DeepSurv but effectively zero for every linear/kernel model (Cox, SVM, Weibull all assign it 0.0000) — the same family of models that underperforms on discrimination.

Dynamic-DeepHit is the best-calibrated model at both 2- and 5-year horizons; Cox is calibrated backwards at 2 years (predicted risk rises while ob-

Model	C-idx↑	Brier↓	AUC↑
Cox	0.288	0.278	0.175
Dynamic-DeepHit	0.850	0.107	0.929
Hazard Transformer	0.190	0.221	0.059
Logistic Hazard	0.373	0.663	0.377
RNN-Surv	0.441	0.287	0.232
DeepSurv	0.653	0.231	0.569
GBSA	0.542	0.220	0.531
Survival RF	0.428	0.293	0.260
Survival SVM	0.585	0.217	0.787
Weibull AFT	0.712	0.226	0.738

Table 3: Discrimination, twenty_features (heterogeneous).

served risk falls) — the calibration-level symptom of its below-chance discrimination. For decision-curve analysis, every model except Cox exceeds the eGFR-threshold referral rule’s net benefit across the clinically relevant 0.05–0.30 threshold range — the clearest evidence here that model-guided referral beats a simple eGFR-cutoff rule already used in practice, and the one result that does not depend on resolving Cox’s open question.

3.3. Twenty Features

Table 3 reports discrimination for the 20-feature heterogeneous scenario (10 models, no KFRE analogue). The same pattern reappears: Dynamic-DeepHit leads by a wide margin, and Cox (and here, more severely, Hazard Transformer) score below chance while every other model stays at or above random — evidence the Cox/Hazard-Transformer issue is not specific to the four/eight-feature merge design, though a common root cause has not been isolated.

4. Discussion

More features did not clearly help. Adding calcium, phosphate, bicarbonate, and albumin to the 4-variable KFRE set left nine of eleven models within 0.03 C-index of their four-feature result. Read with KFRE’s own middling position among the ten learned models, this favors the hypothesis that most recoverable signal in a small, clinically curated feature set is already present at four variables — mirroring KFRE’s original design intuition — with Dynamic-DeepHit’s full-sequence architecture, not feature count, the source of its advantage.

Evaluation unit matters as much as model choice. The reversal between row-level and patient-level evaluation (Section 2.4) is, in our view, the

most transferable finding here, independent of which model wins. Any survival benchmark built on a counting-process row format should confirm, per model, whether test-time scoring happens once per patient or once per row, since the two can rank models differently, as they did here for KFRE and Cox.

Decision-curve analysis is robust to the open Cox question. Regardless of how Cox’s below-chance discrimination is eventually explained, nearly every model exceeds a simple eGFR-threshold referral rule’s net benefit — the clearest actionable finding here, and the one least contingent on that open question.

4.1. Limitations

Cox’s below-chance discrimination is unexplained. It is consistent across all three feature-set scenarios, ruling out a scenario-specific data bug, but we have not isolated whether the cause is model non-convergence, a sign/encoding issue specific to this row format, or something else; Cox’s numbers here should be read as evidence that *something* in this representation is unusually hard for a linear proportional-hazards model, not as a validated statement that Cox is simply worse here. **No competing-risks modeling:** death before ESRD is censored, not modeled as a competing event, since recovering each subject’s absolute calendar anchor was out of this pipeline’s current scope. **Feature importance is model-native, not SHAP:** gradient saliency, permutation importance, and raw coefficients are not on a common scale across model families. Full architectural detail, cohort-merge derivation, and citations for every design choice above are in the Appendix.

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Appendix A. Cohort Construction: Full Derivation

KFRE’s inputs are drawn from separate, asynchronously-ordered labs (serum creatinine, a urine albumin/creatinine test, and, for the 8-variable form, a chemistry panel) — unlike a design that avoids multi-lab merging altogether by encoding one row per single-lab event with missingness flags (the twenty_features scenario’s approach, Section 2.1). Producing real, simultaneous, non-missing covariate rows for four_features/eight_features required a merge design with two open sub-problems: (1) whether the row format should still be time-varying given that KFRE itself is a point-in-time equation, and (2) how to handle a patient having multiple draws of the same lab within one admission.

Row format. We keep the time-varying (`start`, `stop`) format used throughout. KFRE’s formula is agnostic to row packaging — it can be applied independently to each row, treating every row as its own landmark snapshot in the sense of landmarking methodology for time-varying covariates in survival analysis (van Houwelingen, 2007; Putter and van Houwelingen, 2022).

Merge rule: creatinine-anchored, bounded nearest-value windows. Each row is anchored on one creatinine measurement (eGFR is computed from it, via the CKD-EPI 2021 equation, at that instant). Other required labs are attached to the anchor by nearest-value matching within an explicit, literature-backed window rather than by unbounded last-observation-carried-forward, which is known to bias sparse EHR-derived time-varying covariates (Hu et al., 2022):

- *Calcium, phosphate, bicarbonate, serum albumin* (8-variable scenario only, routinely drawn on the same chemistry panel as creatinine): nearest value within ± 24 hours of the anchor. This window follows APACHE II’s precedent of treating all labs drawn within a patient’s first 24 hours — including serum creatinine — as one physiologic snapshot (Knaus et al., 1985); no directly analogous 24h-snapshot convention was found for other candidate windows (e.g., KDIGO’s 48h AKI change-detection window measures a *trend*, not concurrency, and is not equivalent).
- *uACR* (a separately-ordered urine test): nearest value across the patient’s entire history. We initially bounded this match to the same hospital admission, following the resolution Tangri et al. document for one of their own real-world validation cohorts (*“Geisinger: covariates obtained most closely to index date within a past year were included in models”* (Tangri et al., 2016)), but this bound caused 96.5% patient attrition ($34,332 \rightarrow 1,220$ for the 4-feature scenario) and skewed the outcome rate from 91.9% to 98.2% ESRD-positive by disproportionately dropping ESRD-negative patients. We loosened the bound to the patient’s whole history, trading tighter same-encounter simultaneity for a workable, less-biased cohort size — an accepted, documented tradeoff, not an oversight.
- Any row missing a required lab within its window is *dropped*, not imputed or flagged —

four_features/eight_features are complete-case by construction, matching KFRE’s own derivation cohort’s eligibility criteria (Tangri et al., 2011).

One row is emitted per qualifying creatinine draw, so a patient monitored multiple times within one admission correctly yields multiple rows, each independently paired with its own nearest in-window value of every other required lab (a sparser lab such as uACR is therefore legitimately reused, at its nearest value, across several rows for the same patient — expected under the landmarking framework above, not a bug).

Table 1 follows the two-column cohort-flow reporting convention of Major et al. (Major et al., 2019) and STROBE item 13(a) (von Elm et al., 2007). All three scenarios are event-rich by construction (83–92% patient-level ESRD-positive), because the source population is itself drawn from a CKD/ESRD diagnosis-code criterion.

Cohort-size adequacy. eight_features’ 1,213 patients / 1,032 events is the smallest cohort in this study. Benchmarked against the KFRE literature itself, this is not undersized: it exceeds 9 of the 15 published cohorts (263–12,955 patients) on which the 8-variable KFRE was externally validated (Tangri et al., 2016), and its events-per-variable ratio (≈ 129) is roughly $13\times$ the classical $EPV \geq 10$ adequacy floor for stable regression coefficients (Peduzzi et al., 1996). This EPV heuristic directly supports the Cox and KFRE benchmarks; it has no established analogue for the more data-hungry deep architectures also benchmarked here, so eight_features remains the scenario most likely to be data-constrained for those models specifically.

Appendix B. KFRE Coefficients: Full 8-Variable Form

$$L_8 = -0.1992 \left(\frac{\text{age}}{10} - 7.036 \right) + 0.1602(\text{male} - 0.5642) \\ - 0.4919 \left(\frac{\text{eGFR}}{5} - 7.222 \right) + 0.3364(\ln(\text{uACR}) - 5.137) \\ - 0.3441(\text{albumin} - 3.997) + 0.2604(\text{phosphate} - 3.916) \\ - 0.07354(\text{bicarb.} - 25.57) - 0.2228(\text{calcium} - 9.355) \quad (2)$$

Coefficients from Tangri et al. (2011); North-American baseline survival constants from Tangri et al. (2016). KFRE has no published equation for the twenty_features scenario and is therefore absent from it.

Appendix C. Benchmarked Models: Full Architectural Detail

Every neural and ensemble model is retuned per scenario (below), so we describe each model’s structural form and loss precisely, verified directly against the model implementations, and give the tuned hyperparameter range rather than a single fixed value.

C.1. Traditional Survival Analysis Models

Both are fit via `lifelines` (Davidson-Pilon, 2019) directly on the counting-process (`start`, `stop`) rows.

Cox proportional hazards (Cox, 1972) models the hazard of subject i at time t as $\lambda_i(t) = \lambda_0(t) \exp(\mathbf{x}_i(t)^\top \boldsymbol{\beta})$, where $\mathbf{x}_i(t)$ is subject i ’s (possibly time-varying) covariate row and $\lambda_0(t)$ an unspecified baseline hazard, fit by maximizing the Efron-corrected partial likelihood over all subjects’ `start/stop` intervals directly (`lifelines`’ `CoxTimeVaryingFitter`, `id_col=subject_id`) rather than one static row per patient — the model consumes every landmark row (Appendix A), not just each patient’s last. We use an L2 penalizer of 1.0 to stabilize the fit against the strongly collinear rows a single patient contributes.

Weibull Accelerated Failure Time (AFT) (Anderson, 1991) instead parameterizes the full survival distribution, $S(t | \mathbf{x}) = \exp(-(t/\lambda(\mathbf{x}))^\rho)$, $\lambda(\mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\beta})$, fit on the flattened one-row-per-patient (last-observation) frame via `lifelines`’ `WeibullAFTFitter`. For twenty `features` (whose 40, mostly-zero-filled columns produce a near-singular design matrix), we add an L2 penalizer of 0.1 on the scale parameter’s coefficients, the library’s own documented remedy; four `features`/eight `features` use the unpenalized default.

C.2. Deep Survival Analysis Models

All five consume the flattened last-observation frame except Dynamic-DeepHit, which consumes each patient’s full row sequence directly.

DeepSurv (Katzman et al., 2018) is a feedforward network — 1 to 20 `Linear`→`ReLU`→`Dropout` blocks (width 16–256, per-layer dropout 0.1–0.5, all tuned) followed by a single linear output — trained with the Cox partial log-likelihood on its scalar output, so its raw output is already a “higher = riskier” hazard score by construction.

Dynamic-DeepHit (Lee et al., 2020, 2018) feeds each patient’s full covariate/mask/time-gap row se-

quence directly into a 2-layer *bidirectional* LSTM (hidden width 16–128, dropout 0.1–0.5, tuned). The concatenated forward/backward final states pass through a `Linear`→`Tanh` compression, then a masked additive (Bahdanau-style) attention layer pools the sequence into one context vector per patient, letting the model weight informative visits over uninformative ones. The pooled vector feeds 0–3 further cause-specific `Linear`→`ReLU`→`Dropout` layers (width 16–128, tuned) and a final per-risk linear head producing a probability mass function (PMF) over 5,475 daily bins (15 years). Following DeepHit’s original formulation (Lee et al., 2018), training minimizes a composite loss $L = L_1 + \alpha L_2$: L_1 is the negative log-likelihood of the observed interval-censored outcome, and L_2 is a pairwise ranking loss rewarding the model for assigning an event subject higher cumulative incidence, at that subject’s own event time, than any subject not yet failed by then — with $\alpha \in [0.1, 1.0]$ tuned per scenario.

RNN-Surv (Giunchiglia et al., 2018) embeds each timestep through 1–3 `Linear`→`ReLU` layers (width 32–128, tuned) into a 1–3-layer *LSTM* (hidden width 64–256, tuned). A final linear layer maps the last hidden state to logits over $K \in [10, 50]$ (tuned) discrete intervals spanning the 730-day horizon, softmax-normalized into a PMF. As with Dynamic-DeepHit, the loss combines a discrete-time likelihood (observed-interval event probability for events; survival-beyond-interval probability for censored subjects) with a pairwise ranking term over the cumulative incidence function, weighted by a tunable coefficient in $[0.1, 0.9]$.

LogisticHazard discretizes the 730-day horizon into 50 quantile-based bins (`pycox`’s `LabTransDiscreteTime`) and learns the log-odds of failure in each bin, via `pycox`’s off-the-shelf `MLPVanilla` network (1–3 hidden layers, width $\in \{16, 32, 64, 128\}$, dropout 0–0.5, all tuned, batch size $\in \{32, 64, 128, 256\}$) wrapping its `LogisticHazard` discrete-time hazard head and cross-entropy-style loss.

Hazard Transformer (Oliver et al., 2024; Vaswani et al., 2017) departs from a sequence-attention reading of “Transformer”: each patient’s rows are linearly embedded and *mean-pooled* (masked over observed rows) into a single patient representation *before* any attention is applied. This pooled vector is broadcast across 100 fixed query time points spanning $[0, 730]$ days, each combined with a learned time-encoding and a sinusoidal positional encoding;

2–6 (tuned) Transformer encoder layers (1–8 heads, tuned) then self-attend *across these 100 time queries*, not across the patient’s visits. A per-risk decoder produces a softmax-normalized PMF over the 100 bins (an earlier per-bin independent-sigmoid parametrization collapsed to a constant prediction and was replaced by this bounded-PMF form, the same fix Dynamic-DeepHit’s risk heads use). Training uses the same composite likelihood-plus-ranking loss family as above.

C.3. Ensemble-Learning-Based Models

All three are `scikit-survival` (Pölsterl, 2020) estimators fit on the flattened last-observation frame.

Gradient Boosting Survival Analysis (GBSA) (Chen et al., 2013) fits an additive ensemble of regression trees against the Cox partial-likelihood gradient; grid-searched over `n_estimators` ∈ {50, 100, 200, 300}, `max_depth` ∈ {3, 5, 10, 15}, `learning_rate` ∈ {0.01, 0.1, 0.2} by cross-validated C-index.

Random Survival Forest (SRF) (Pickett et al., 2021; Ishwaran et al., 2008) grows an ensemble of survival trees splitting on the log-rank statistic; grid-searched over `n_estimators` ∈ {50, 100, 150}, `max_depth` ∈ {None, 5, 10}. Since `scikit-survival` does not expose impurity-based feature importance for this estimator (known to be biased for survival forests), we compute permutation importance for it directly.

FastSurvivalSVM learns a linear ranking function over the covariate space, optimizing a convex upper bound on the concordance index subject to margin and ℓ_2 constraints. Fit with `scikit-survival`’s defaults ($\alpha = 0.01$), the one model here run without any hyperparameter search.

C.4. Model Training and Optimization

Every tuned range above (DeepSurv, Dynamic-DeepHit, RNN-Surv, LogisticHazard, Hazard Transformer) is searched via 10-trial Bayesian search (Optuna) per scenario, selecting the trial with best validation loss; GBSA/SRF use grid search selected by cross-validated C-index. FastSurvivalSVM, Cox, and Weibull AFT use fixed, untuned hyperparameters. All neural models use Adam with early stopping on a held-out validation split.